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Chatbot to chat with medical books using Retrieval-Augmented Generation Model



Team
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Contents

1. Abstract
2. Introduction
3. Literature Survey
4. Problem Identification
5. Proposed System
6. Software and Hardware Requirements
7. Project Plan
8. Current Work
9. Conclusion



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Abstract

MEDBUDDY: AI-Powered Medical Chatbot using Retrieval-Augmented Generation

Generic AI hallucinations present a critical risk in medical education, as standard models often lack transparency regarding their sources and are prone to providing false syllabus information. To address this, Med-Buddy is presented as an intelligent Retrieval-Augmented Generation (RAG) framework designed to create a secure, hallucination-free study environment by mathematically matching user queries to verified textbook paragraphs. While the base implementation utilized the Google Gemini API within the LlamaIndex framework, it faced significant operational hurdles due to restrictive rate limits and high costs; therefore, our primary contribution involves transitioning the architecture to the Grok LLM to effectively eliminate API exhaustion issues while integrating TruLens for advanced RAG evaluation—utilizing "RAG Triad" metrics (context relevance, groundedness, and answer relevance) to provide real-time quality scores. Furthermore, the framework was ehampted is its caten for. Furthermore, the framework was enhanced by integrating Local FastEmbed for on-device vectorization and a perɵisistent indexing mechanism using MD5 hashing to allow interrupted document processing to resume seamlessly, all deployed via a Streamlit-based user interface that offers a responsive, web-based dashboard for seamless interaction with verified medical literature. By combining the high-speed Grok engine with optimized local data parsing and rigorous metric-based validation, Med-Buddy offers a scalable, cost-effective solution that transforms unstructured medical literature into actionable, verified insights for students.

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Introduction

- The rising demand for skilled professionals necessitates better preparation tools. To bridge the gap in resources and empower individuals, our project develops an AI-powered platform.
- This application is designed to tailor questions and insights to the unique requirements of the user's focus area.
- The platform offers valuable feedback on user responses and the correct answers of the questions based strictly on provided documentation.
- By offering a comprehensive and personalized preparation experience, we help users refine their skills, boost their confidence, and ultimately succeed in their academic and career pursuits.



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Literature Survey

Sl No.	Title [Year]	Author	Proposed Method	Disadvantages
1	AI Chatbots in Medical Education and Public Health [2023]	Francesco Baglivo et al., Gunther Eysenbach, Taha Bin Arif et al.	Uses ChatGPT to support vaccination training, enhance learning, and assist research.	Lacks real-world implementation analysis; issues with misinformation and ethical concerns.
2	AI Chatbots for Specialized Medical Training [2022]	Jeong-Won Han et al., David Kovacek & James C L Chow	Interactive learning for nursing and radiation safety to improve engagement.	Lacks scalability analysis; does not cover integration or adaptability issues.
3	AI Chatbots with Biomedical and Data Query Systems [2018/2023]	V. Murali et al., R. S. Nambiar	Uses ChEMBL Bot and BERT frameworks to improve responses via NLP.	Requires domain-specific tuning; limited by contextual understanding and dataset dependency.

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Literature Survey

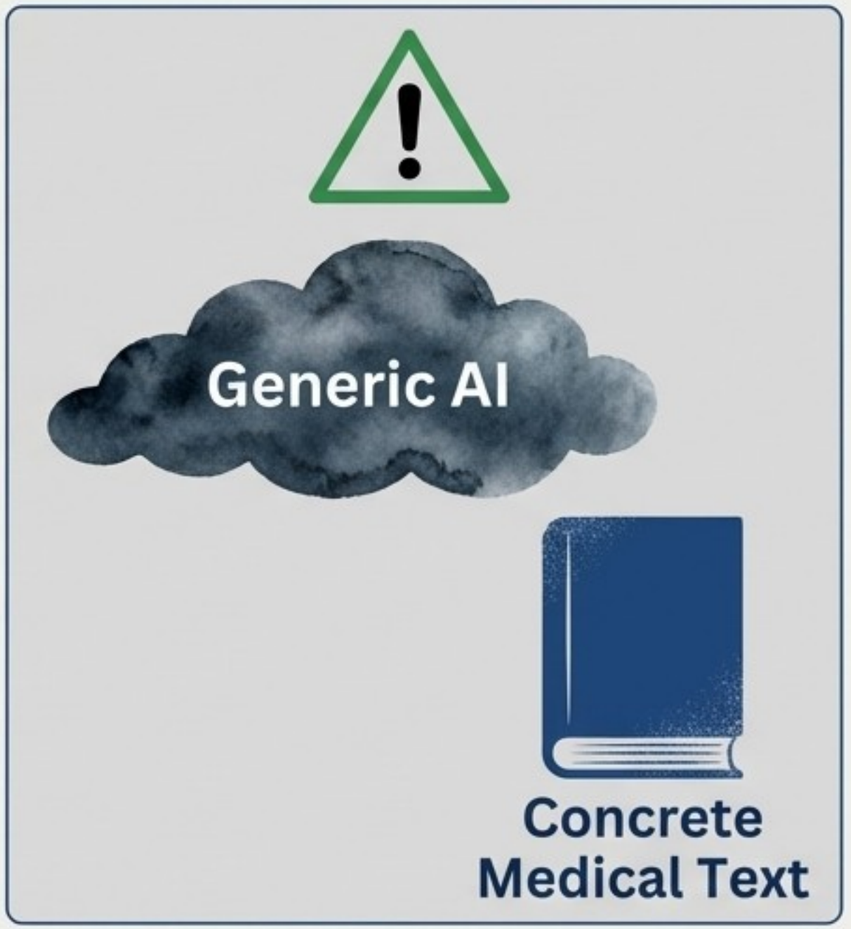
SI No.	Title [Year]	Author	Proposed Method	Disadvantages
4	Deep Learning and Ontology-Based Chatbot Systems [2022/2023]	P. C. Nair, Geethalekshmy et al.	Explains neural networks and semantic reasoning for intelligence and personalization.	High computational complexity; limited emotional intelligence and real-world validation.
5	Retrieval-Augmented Generation (RAG) in Medical Chatbots [2024]	Sai Deepa Bhavani Peri et al.	Combines retrieval and generation for fast querying of medical textbooks.	Performance depends heavily on data quality; risk of incomplete answers if input is convictent; ronics answers if input is insufficient.

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Problem Identification

- **Lack of Transparency:** Information provided by normal generic AI is neither transparent nor concrete about the source of the information.
- **The Hallucination Risk:** Generic models use LLMs to generate answers which have a high possibility of hallucination and cannot provide a particular source for those answers.
- **Critical Consequences:** Students become vulnerable to false information or data not related to their syllabus. Misleading medical students with wrong information is a critical failure that might cost lives in the future.



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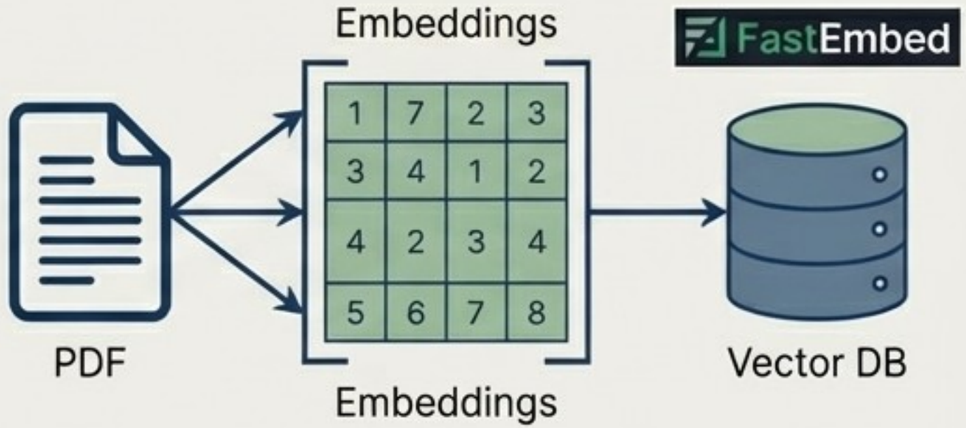
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Proposed System

MedBuddy Overview: Uses the high-speed Grok LLM combined with Local FastEmbed to ensure accuracy, eliminate API exhaustion, and mathematically match uploaded PDFs.

Key Concepts:

- **Embeddings** (Local FastEmbed): The process of turning human words and sentences into a list of numbers so the computer can mathematically understand their
- **Vector Database** (Persistent Indexing): A special storage space utilizing MD5 hashing to keep these 'number-words,' allowing the AI to instantly resume processing and find
- **LLM Grounding** (TruLens Evaluated): A strict rule placed on the Grok engine, evaluated in real-time by TruLens, forcing it to only generate an answer using the text to only generate an answer using the text provided.



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MedBuddy System Architecture: The Engine Room

User Interface

Streamlit



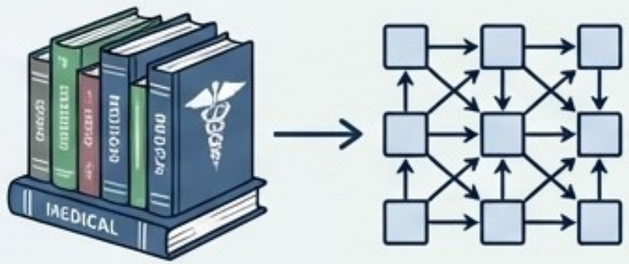
Streamlit Web App: Seamless, real-time query input via an interactive dashboard.

API Requests / Chat Responses

RAG Pipeline & Data Management

Local FastEmbed & MD5 Hashing

Persistent Indexing: Local FastEmbed for on-device vectorization and MD5 hashing for resume-able document processing.



Raw Medical PDFs → Vector Embeddings.

Context Ingestion & Prompt Execution

Generative Engine

Grok LLM: High-speed reasoning, eliminating rate limits and API exhaustion.

Grounding Verification

Grounding & Safety

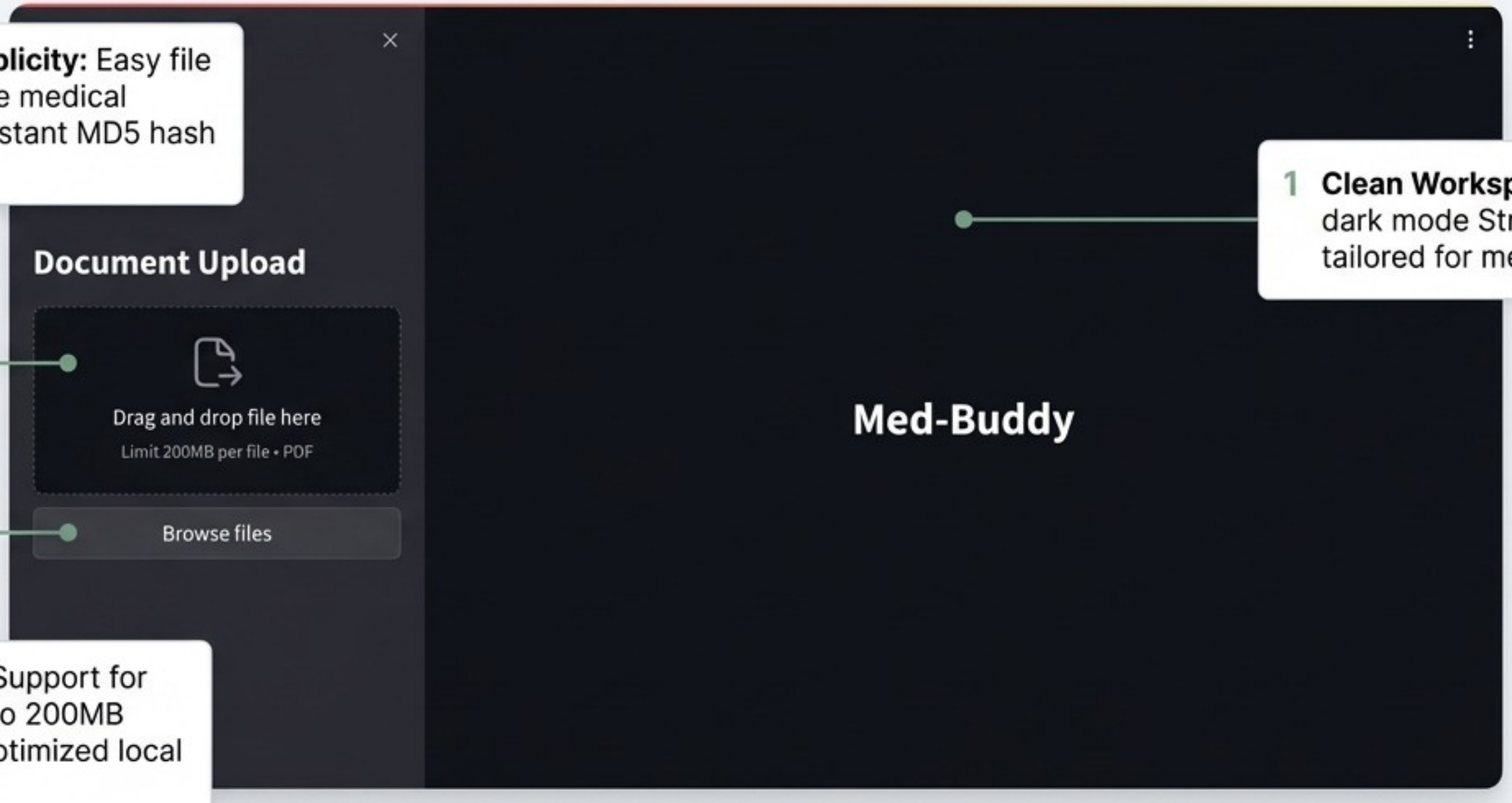
TruLens

TruLens Evaluation: Real-time "RAG Triad" scoring (context relevance, groundedness, answer relevance).

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A frictionless interface for secure medical document upload

2 **Drag & Drop Simplicity:** Easy file intake for massive medical textbooks with instant MD5 hash checking.

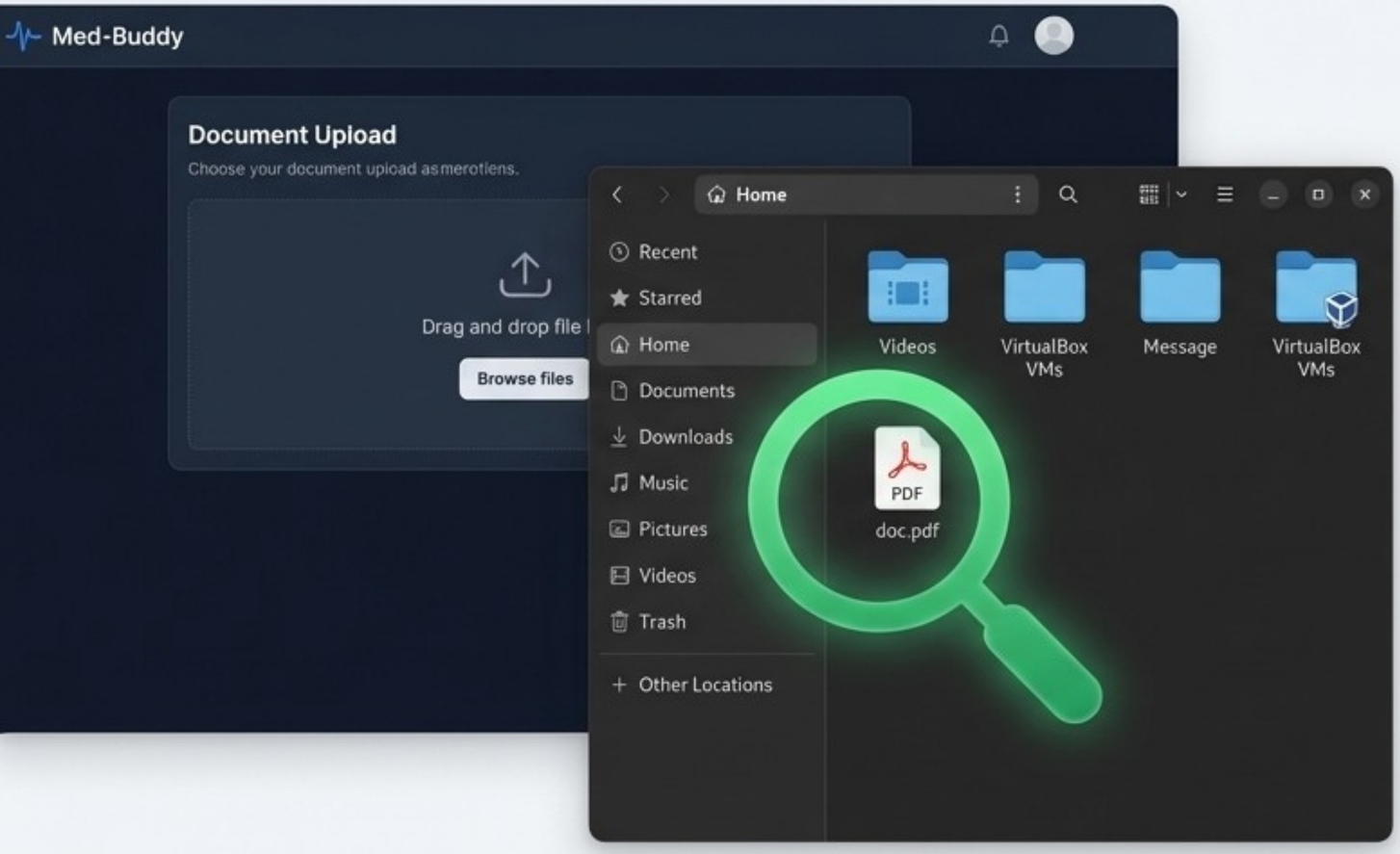


1 **Clean Workspace:** Distraction-free dark mode Streamlit environment tailored for medical focus.

3 **High Capacity:** Support for dense PDFs up to 200MB processed via optimized local data parsing.

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Seamless bridging between local environments and the web application



Users can instantly map local medical literature (e.g., standard curriculum PDFs) directly into the secure Med-Buddy indexing pipeline.

By utilizing Local FastEmbed, **files are vectorized on-device without cloud exposure risks, ensuring zero API token costs during ingestion.**

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A focused chat environment restricted strictly to verified contexts

1 **The Learning Zone:** Once uploaded, the Grok engine is securely locked to the document context, actively monitored by TruLens guardrails.



2 **Instant Retrieval:** Queries entered here bypass generic web data, utilizing the high-speed Grok API to fetch only mathematically matched textbook chunks.

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Conclusion

Med-Buddy successfully provides a safe, grounded, and hallucination-free AI assistant. By replacing standard APIs with the high-speed Grok engine, Local FastEmbed, and TruLens validation, it achieves a highly scalable, cost-effective solution. It democratizes access to complex medical knowledge, allowing students to study smarter.

Future Scope

Fully Localized Inference.
Transitioning entirely to locally hosted LLMs on hospital hardware to remove final cloud steps.

Multi-Modal RAG.
Expanding ingestion to analyze medical imagery (X-rays, MRIs) alongside textual textbooks.

EHR Integration.
Utilizing HL7/FHIR standards to cross-reference patient electronic health records against uploaded guidelines.

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References

1. F. Baglivo et al., "AI Chatbots in Medical Education and Public Health," Frontiers in Medicine, 2023.
2. S. D. B. Peri et al., "Retrieval-Augmented Generation (RAG) in Medical Chatbots," IEEE International Conference, 2024.
3. P. C. Nair et al., "Deep Learning and Ontology-Based Chatbot Systems," International Conference on Inventive Computation Technologies, 2022.
4. J. Han et al., "AI Chatbots for Specialized Medical Training," JMIR Medical Education, 2022.
5. V. Murali et al., "ChEMBL Bot - A Chat Bot for ChEMBL Database," International Conference on Information Processing, 2018.
6. TruLens Documentation, "RAG Triad Evaluation Metrics," 2024. Retrieved from trulens.org.

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